Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112115
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 26 Question Id : 691121176 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Dimensions of universal gravitational constant (G) in terms of Planck's constant (h), distance (L), mass (M) and time (T) are _____.

Options :

691121596. $[hTLM^{-2}]$

691121597. $[hT^{-1}LM^{-2}]$

691121598. $[hTL^{2}M^{-2}]$

691121599. $[h^{-1}T^{-1}LM^{-2}]$

Question Number : 27 Question Id : 691121177 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A 0.5 kg mass is in contact against the inner wall of a cylindrical drum of radius 4 m rotating about its vertical axis. The minimum rotational speed of the drum to enable the mass to remain stuck to the wall (without falling) is 5 rad/s. The coefficient of friction between the drum's inner wall surface and mass is _____. (Take $g = 10\ m/s^2$)

Options :

691121600. 0.1

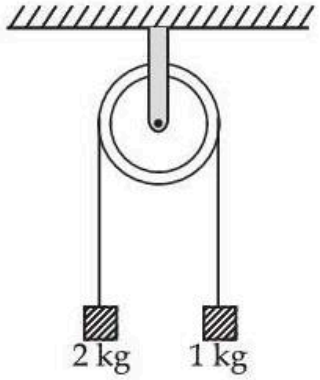
691121601. 0.5

691121602. 0.7

691121603. 0.3

Question Number : 28 Question Id : 691121178 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two blocks of masses 2 kg and 1 kg respectively, are tied to the ends of a string which passes over a light frictionless pulley as shown in the figure below. The masses are held at rest at the same horizontal level and then released. The distance traversed by the centre of mass in 2 s is _____m. (Take $g = 10\ m/s^2$)



Options :

691121604. 3.33

691121605. 3.12

691121606. 2.22

691121607. 1.42

Question Number : 29 Question Id : 691121179 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A particle having charge 10^{-9} C moving in x-y plane in fields of $0.4\ \hat{j}$ N/C and $4 \times 10^{-3}\ \hat{k}$ T experiences a force of $\left(4\hat{i} + 2\hat{j}\right) \times 10^{-10}$ N. The velocity of the particle at that instant is _____ m/s.

Options :

691121608. $50\hat{i} + 100\hat{j}$

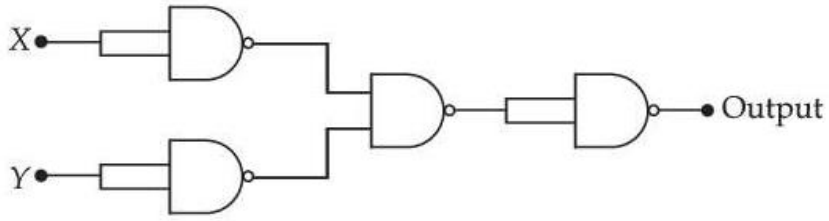
691121609. $100\hat{i} + 50\hat{j}$

691121610. $-50\hat{i} + 100\hat{j}$

691121611. $50\hat{i} - 100\hat{j}$

Question Number : 30 Question Id : 691121180 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If X and Y are the inputs, the given circuit works as _____.



Options :

691121612. OR gate

691121613. AND gate

691121614. NAND gate

691121615. NOR gate

Question Number : 31 Question Id : 691121181 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If a body of mass 1 kg falls on the earth from infinity, it attains velocity (v) and kinetic energy (k) on reaching the surface of earth. The values of v and k respectively are _____ .
(Take radius of earth to be 6400 km and $g = 9.8\ m/s^2$)

Options :

691121616. 11.2 km/s; 6.27×10^7 J

691121617. 11.2 km/s; 12.54×10^7 J

691121618. 8.8 km/s; 6.27×10^7 J

691121619. 8.8 km/s; 12.54×10^7 J

Question Number : 32 Question Id : 691121182 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In a screw gauge the zero of main scale reference line coincides with the fifth division of the circular scale when two studs are in contact. There are 100 divisions in circular scale and pitch of screw gauge is 0.1 mm. When diameter of a sphere is measured, the reading of main scale is 5 mm and 50th division of circular scale coincides with the reference line of main scale. The diameter of sphere is _____ mm.

Options :

691121620. 5.045

691121621. 5.055

691121622. 5.450

691121623. 5.550

Question Number : 33 Question Id : 691121183 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The surface tension of a soap bubble is 0.03 N/m. The work done in increasing the diameter of bubble from 2 cm to 6 cm is $\alpha\pi \times 10^{-4}$ J. The value of α is _____. (Take $\pi = 3.14$)

Options :

691121624. 0.86

691121625. 0.64

691121626. 1.92

691121627. 7.68

Question Number : 34 Question Id : 691121184 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A mixture of carbon dioxide and oxygen has volume 8310 cm^3, temperature 300 K, pressure 100 kPa and mass 13.2 g. The number of moles of carbon dioxide and oxygen gases in the mixture respectively are _____.

(Assume both carbon dioxide and oxygen gases behave like ideal gases) [R = 8.31 J/mol.K]

Options :

691121628. 0.15 and 0.18

691121629. 0.25 and 0.08

691121630. 0.21 and 0.12

691121631. 0.13 and 0.20

Question Number : 35 Question Id : 691121185 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If an air bubble of diameter 2 mm rises steadily through a liquid of density 2000 kg/m^3 at a rate of 0.5 cm/s, then the coefficient of viscosity of liquid is _____ Poise. (Take $g = 10\ m/s^2$)

Options :

691121632. 0.88

691121633. 8.8

691121634. 88.8

691121635. 0.088

Question Number : 36 Question Id : 691121186 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A spherical ball of mass 2 kg falls from a height of 10 m and is brought to rest after penetrating 10 cm into sand. The average force exerted by sand on the ball is _____ N.
(Take $g = 10\ m/s^2$)

Options :

691121636. 1980

691121637. 2020

691121638. 2000

691121639. 1000

Question Number : 37 Question Id : 691121187 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An electromagnetic wave travels in free space along the x-direction. At a particular point in space and time, $\vec{B} = 2\times10^{-7}\ \hat{j}$ T is associated with this wave. The value of corresponding electric field $\vec{E}$ at this point is _____ V/m.

Options :

691121640. $60\ \hat{k}$

691121641. $-60\ \hat{k}$

691121642. $30\ \hat{k}$

691121643. $-600\ \hat{k}$

Question Number : 38 Question Id : 691121188 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two resistors of 200 Ω and 400 Ω are connected in series with a battery of 100 V. A bulb rated at 200 V, 100 W is connected across the 400 Ω resistance. The potential drop across the bulb is _____V.

Options :

691121644. 25

691121645. 50

691121646. 66.6

691121647. 100

Question Number : 39 Question Id : 691121189 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two metal plates *(A, B)* are kept horizontally with separation of $\left(\frac{12}{\pi}\right)$ cm, with plate A on the top. An atomizer jet sprays oil (density 1.5 g/cm^3) droplets of radius 1 mm horizontally. All oil droplets carry a charge 5 nC. The potentials V_A and V_B are required on plates A and B respectively in order to ensure the droplets do not descend. The values of V_A and V_B are _____.
(Neglect the air resistance to the droplets and take $g = 10\ m/s^2$)

Options :

691121648. 100 V and 580 V

691121649. 580 V and 100 V

691121650. 60 V and 400 V

691121651. 0 V and $-200\ V$

Question Number : 40 Question Id : 691121190 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two point charges 8 μC and $-2\ \mu C$ are located at $x = 2$ cm and $x = 4$ cm, respectively on the x-axis. The ratio of electric flux due to these charges through two spheres of radii 3 cm and 5 cm with their centers at the origin is _____.

Options :

691121652. 4 : 1

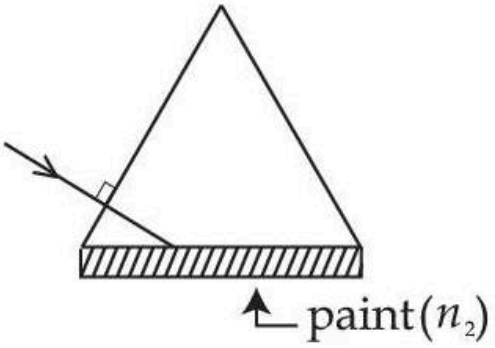
691121653. 3 : 4

691121654. 4 : 3

691121655. 4 : 5

Question Number : 41 Question Id : 691121191 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

One side of an equilateral prism is painted by a transparent material of refractive index n_2. The refractive index of prism is 1.6. The minimum value of n_2 required for total internal reflection from painted face is _____.



Options :

691121656. $3\sqrt{3}/1.6$

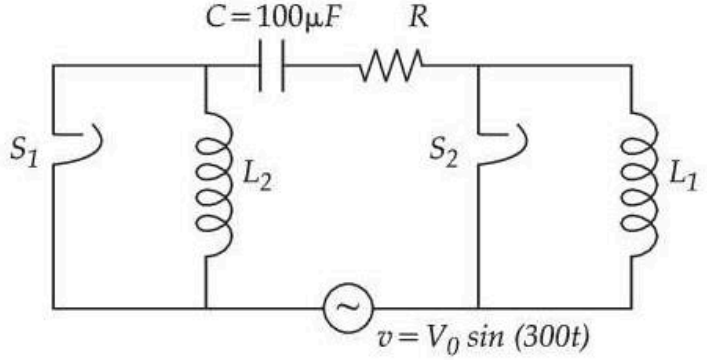
691121657. $\sqrt{3}$

691121658. $3.2/\sqrt{3}$

691121659. $4\sqrt{3}/5$

Question Number : 42 Question Id : 691121192 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The figure given below shows an *LCR* series circuit with two switches S_1 and S_2. When switch S_1 is closed keeping S_2 open, the phase difference (ϕ) between the current and source voltage is 30° and phase difference is 60° when S_2 is closed keeping S_1 open. The value of $(3L_1 - L_2)$ is _____ H.



Options :

691121660. $\frac{9}{2}$

691121661. $\frac{2}{9}$

691121662. $\frac{1}{3}$

691121663. 3

Question Number : 43 Question Id : 691121193 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A circular current loop of radius R is placed inside square loop of side length L ($L >> R$) such that they are co-planar and their centers coincide. The permeability of free space is μ_0. The mutual inductance between circular loop and square loop is _____.

Options :

691121664. $2\sqrt{2}\frac{\mu_0 L^2}{R}$

691121665. $\sqrt{2}\frac{\mu_0 L^2}{R}$

691121666. $\sqrt{2}\frac{\mu_0 R^2}{L}$

691121667. $2\sqrt{2}\frac{\mu_0 R^2}{L}$

Question Number : 44 Question Id : 691121194 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The binding energy per nucleon of $^{209}_{83}Bi$ is _____ MeV.

[Take $m\left(^{209}_{83}Bi\right) = 208.980388$ u, $m_p = 1.007825$ u, $m_n = 1.008665$ u, 1 u = 931 MeV/c^2]

Options :

691121668. 7.48

691121669. 7.84

691121670. 8.79

691121671. 6.94

Question Number : 45 Question Id : 691121195 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The equation of motion of a particle is given by $x = a\sin\left(50t + \pi/3\right)$ cm. The particle will come to rest at time t_1 and it will have zero acceleration at time t_2. The t_1 and t_2 respectively are _____.

Options :

691121672. $\frac{\pi}{300}\,s, \frac{\pi}{75}\,s$

691121673. $\frac{\pi}{75}\,s, \frac{\pi}{300}\,s$

691121674. $\frac{\pi}{300}\,s, \frac{\pi}{25}\,s$

691121675. $\frac{\pi}{50}\,s, \frac{\pi}{100}\,s$

Physics Section B

Section Id :	69112116
Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112116
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 46 Question Id : 691121196 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

In a Young's double slit experiment, the intensity at some point on the screen is found to be $\frac{3}{4}$ times of the maximum of the interference pattern. The path difference between the interfering waves at this point is $\frac{\lambda}{x}$ where λ is wavelength of the incident light. The value of x is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 47 Question Id : 691121197 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Using Bohr's model, calculate the ratio of the magnetic fields generated due to the motion of the electrons in the 2nd and 4th orbits of hydrogen atom _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 48 Question Id : 691121198 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

5 moles of unknown gas is heated at constant volume from 10 °C to 20 °C. The molar specific heat of this gas at constant pressure $c_p = 8$ cal/mol. °C and $R = 8.36$ J/mol.°C. The change in the internal energy of the gas is _____ calorie.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

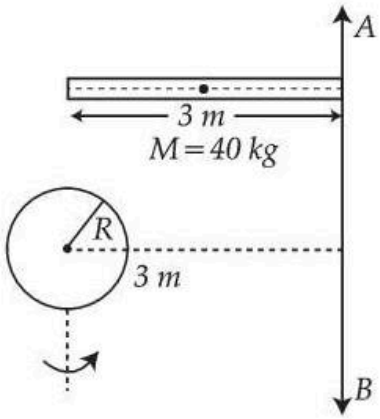
Question Number : 49 Question Id : 691121199 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If sunlight is focused on a paper using convex lens, it starts burning the paper in shortest time when the lens is kept at 30 cm above the paper. If the radius of curvature of the lens is 60 cm then the refractive index of the lens material is $\frac{\alpha}{10}$. The value of α is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 50 Question Id : 691121200 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Moment of inertia about an axis AB for a rod of mass 40 kg and length 3 m is same as that of a solid sphere of mass of 10 kg and radius R about an axis parallel to AB axis with separation of 3 m as shown in figure below. The value of R is given as $\sqrt{\frac{\alpha}{2}}$. The value of α is _____.



Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Chemistry Section A

Section Id : 69112117